MEHBUR RAHMAN

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EDUCATION

Bachelor of Science in Computer Science & Engineering (CSE) Expected Gr

North South University (NSU)

Relevant Coursework: Data Structures, OOP, Database Systems, Linux, AI, Computer Networking (Cisco Packet Tracer labs).

Higher Secondary Certificate (HSC)

Adamjee Cantonment College | GPA: 5.00

Secondary School Certificate (SSC)

Adamjee Cantonment Public School | GPA: 5.00

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, PHP, SQL
Backend & Web: Flask, React, REST APIs, SQLAlchemy, JWT
Databases: PostgreSQL, MySQL, ChromaDB
Tools & Infrastructure: Git, GitHub, Linux, Docker, Docker Compose, Pytest, Cisco Packet Tracer
AI & Systems: PyTorch, Machine Learning, RAG, Vector Databases

PROJECTS

Privacy-Aware Adaptive Federated Routing (PAFR) | *Tech: Federated RAG, permission-aware routing* **Ongoing Research**

- Investigating selective, permission-aware routing and minimal-disclosure answers across departments; planning synthetic-data comparisons.

FastBus | *Tech: Flask, PostgreSQL, SQLAlchemy, JWT, Docker Compose* **GitHub**

- Engineered the backend for a full-stack bus ticket booking application with live seat availability and admin dashboards.
- Designed database schemas, implemented JWT authentication, and ensured robust deployment using automated migrations.

SkillForge | *Tech: Python, Qwen2.5-Coder-7B, ChromaDB, Local LLMs* **Ongoing Project**

- Developing a self-evolving, fully offline AI coding agent capable of planning, executing, and repairing code within a sandbox.
- Integrated a local Qwen2.5 model and ChromaDB to crystallize verified solutions into a reusable skill memory.

Voice Command and Speaker Recognition | *Tech: Python, PyTorch, Whisper, Flask, SVM* **GitHub**

- Developed a speech intelligence application utilizing a custom PyTorch CNN for commands and a linear SVM for speaker identification.
- Built Flask JSON APIs supporting browser recording and integrated Whisper Base for a general transcription baseline.

South-Stack | *Tech: React, TypeScript, Vite, WebContainer API* **GitHub**

- Collaborated on an AI-powered browser IDE providing a complete virtual filesystem for a "Plan - Execute - Verify" coding loop.
- Spearheaded the vision model provider and image-to-design workflow for seamless user interface generation.

EXPERIENCE & LEADERSHIP

Founder | ShaftX **2025 - Present**

- Import original branded footwear from the UK and manage retail operations, sales, logistics, and distribution within Bangladesh.

Volunteer, Logistics & Operations | IEEE NSU Student Branch

Assistant General Secretary, Indoor Events | ACC Explorer's Club **2018 - 2020**

CERTIFICATIONS & CO-CURRICULAR

- **Certifications:** Excel Boot Camp Certificate (4 modules, 2.5 hrs/module) – Career & Placement Center, North South University
- **Competitive Programming:** Active on Codeforces focusing on data structures, algorithms, and implementation.